



OPEN Automatic 3D railroad alignment detection using modified Hough transform

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In railroad infrastructure, timely and accurate maintenance is essential to ensure both safety and operational efficiency. Rail alignment is a critical element for constructing a reliable digital model of railroads. However, conventional rail alignment detection often relies on manual processes that are subjective and prone to error. With the advent of digital modeling technologies, there is an opportunity to improve the efficiency and accuracy of railroad maintenance. In this study, we propose an automated method for detecting railroad alignment directly from 3D point cloud data using deep learning and computer vision. The method was validated on the Osong railroad test track, achieving an average RMSE of 3.57 mm in rail alignment detection. The proposed approach significantly reduces the time required to construct digital railroad models and enhances the efficiency of railroad maintenance.

Keywords Deep learning, Hough transform, Point cloud data, Railroad alignment, Semantic segmentation

Recent incidents have highlighted a troubling frequency of railroad accidents, which have the potential to inflict catastrophic damage on both human life and infrastructure^{1,2}. Investigations into these accidents have identified a myriad of contributing factors, yet poor rail maintenance consistently emerges as a primary culprit^{3,4}. The imperative to prevent such accidents underscores the necessity of rigorous railroad infrastructure maintenance. However, this task is daunting due to the extensive nature of railroad networks and the continuous operation required by the transportation sector. Consequently, performing safety diagnostics and maintenance interventions poses a considerable challenge. These activities frequently lead to substantial financial losses due to operational downtime. This context necessitates the exploration and adoption of more efficient methodologies for railroad maintenance, aiming to enhance safety while minimizing economic impact.

The advent of recent technologies has significantly broadened the horizons for enhancing maintenance efficiency within various sectors, including the rail industry. A plethora of studies have demonstrated the application of advanced technologies such as sensors, computer vision, and deep learning to structural maintenance challenges^{5–15}. Specifically, in the realm of railroad maintenance, innovative approaches have been explored to improve safety and efficiency. Within the rail industry, Marino et al.¹⁶ introduced a study aimed at real-time detection and inspection of railroad bolts. Hodge et al.¹⁷ proposed a railroad monitoring approach utilizing a wireless sensor network. Mukojima et al.¹⁸ presented a method for detecting railroad obstacles through image processing technology. Flamini et al.¹⁹ suggested the use of drones for monitoring railroad infrastructure. Núñez et al.²⁰ and Jamshidi et al.²¹ introduced methods for railroad maintenance that leverage big data. Santur et al.²² proposed a method for identifying defects on the rail surface using a laser camera. Lee et al.²³ introduced a method for classifying point clouds related to railroad structures using deep learning. Ghiasi et al.²⁴ and Ni et al.²⁵ conducted studies on rail track defect detection. Furthermore, several studies have applied Building Information Modeling (BIM), a digital modeling approach, to railroads^{26–28}.

The creation of an accurate digital model of a railroad infrastructure necessitates precise geometric data, including the alignment of the rails. Initial models are often derived from design-stage drawings. However, these models can be compromised by distortions due to wear and tear or inaccuracies stemming from construction imperfections, leading to significant errors in the digital representation. Furthermore, the absence of design drawings for some structures exacerbates the challenge of obtaining accurate alignment information. To overcome these limitations and accurately capture the as-is geometry of railroad infrastructure, Light Detection and Ranging (LiDAR) technology has become a pivotal tool. LiDAR scanning facilitates the generation of a 3D point cloud that accurately represents the railroad's geometry. Despite its advantages, extracting rail alignment from point cloud data typically requires manual expert intervention. This process is complex, labor-intensive,

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and inherently subjective, leading to potential inconsistencies depending on the operator. This subjectivity underscores the need for more objective and efficient methodologies in the detection and analysis of rail alignment from LiDAR-generated data.

Therefore, in this study, we introduce a new method designed to automatically detect railroad alignment from 3D point cloud data, utilizing deep learning and computer vision. The proposed method is structured into two primary phases: Phase 1, Rail Segmentation, and Phase 2, Rail Alignment Detection. Each phase is meticulously designed to ensure the accurate extraction and analysis of rail alignment data from complex 3D point cloud environments. The detailed methodologies and procedural steps for each phase are outlined in Chap. 3 of this paper. To validate the performance of the automatic rail alignment detection method, experiments were conducted at the Osong railroad test track in Nojang-ri, Jeondong-myeon, Sejong-si, Republic of Korea. The results of these experiments are presented in Chap. 4.

The main contribution and the novelty of the proposed method is described below.

1. Development of automatic railroad alignment detection method using deep learning and computer vision.
2. Development of rail semantic segmentation network using 3D point cloud data.
3. Development of “rail alignment detection” method using segmented rail data.
4. Development of “curved rail alignment detection” method which can automatically detect the alignment of the railroad even in the curved rail.
5. Full-scale experiments of the proposed method using on-site railroad data.

Background

A point cloud is a digital data structure that represents objects as a collection of points in three-dimensional space. Each point contains information such as coordinates and color; however, it lacks semantic meaning, making additional post-processing essential for practical applications. Traditional manual classification methods, while accurate, are both time-consuming and costly. To overcome these limitations, recent advancements have leveraged deep learning techniques.

Maturana et al.²⁹ introduced VoxNet, a pioneering voxel-based framework that enriches 3D point cloud data with semantic information through classification. Although effective in providing semantic insights, VoxNet suffers from computational inefficiency due to the redundant processing of empty voxels, which substantially increases the computational load. To address this issue, Wu et al.^{30,31} proposed SqueezeSeg, a method that reduces computational demands by projecting point clouds into two-dimensional space for classification. Although this approach significantly reduces computational cost compared to voxel-based methods, dimensionality reduction introduces challenges in maintaining classification accuracy. This highlights an inherent trade-off between computational efficiency and semantic precision.

Qi et al.³² made a significant contribution to point cloud data processing with the development of PointNet, a framework designed to provide both cost-effectiveness and high accuracy in point cloud classification. PointNet achieves this by employing max-pooling, a symmetric function that extracts global features by selecting the maximum value across dimensions, thereby ensuring consistent results regardless of point order. To enhance the model's capacity for capturing point-level interactions, PointNet integrates a multilayer perceptron (MLP) to detect features at the individual point level. This allows for detailed analysis of point characteristics while maintaining computational efficiency. To address the challenge of transformation invariance—a common issue caused by variations in orientation and position—PointNet incorporates T-Net, a mini-network that estimates an affine transformation matrix to normalize the point cloud, ensuring robustness to spatial variations.

Despite its innovative design, PointNet has limitations. Because PointNet relies on max-pooling, it retains prominent global features but may overlook subtle local ones. As a result, valuable local information can be lost. To overcome this, Qi et al.³³ proposed PointNet++, which improves upon the original framework by capturing local features through hierarchical grouping. Unlike PointNet's single-stage max-pooling, PointNet++ extracts features at multiple scales, thereby enriching the representation of local structures and enhancing classification accuracy.

A key advancement introduced in PointNet++ is its strategy for addressing the challenge of density imbalance in point clouds. This issue, inherent to 3D data acquisition methods, can result in inconsistencies in model performance. To mitigate this, PointNet++ employs Farthest Point Sampling (FPS), a technique that selectively samples points to ensure a more uniform distribution across the dataset. This strategic sampling improves the model's generalization across varying point densities, leading to enhanced performance in both classification and segmentation tasks. By integrating these methodologies, PointNet++ achieves superior performance compared to its predecessor. Figure 1 presents a detailed visualization of these improvements and structural innovations.

Methodology

Figure 2 provides an overview of the proposed Automatic Railroad Alignment Detection method. This method autonomously extracts the railroad alignment from point cloud data acquired by LiDAR technology. The proposed method is structured into two main phases:

Phase 1 involves the separation and detection of points representing the rail within the entire point cloud. Initially, PointNet++, as introduced in Chap. 2, is applied to segment the rails from the point cloud. Subsequently, the M-estimator Sample Consensus (MSAC) algorithm³⁴ is utilized to remove noise and enhance the accuracy of rail detection.

Phase 2 is focused on the estimation of the alignment. The Hough transform is employed to identify straight alignments within the rail point cloud. To improve the accuracy of alignment detection in curved railroads, this study introduces modifications to the Hough transform. Detailed procedures for each phase are elaborated in the subsequent.

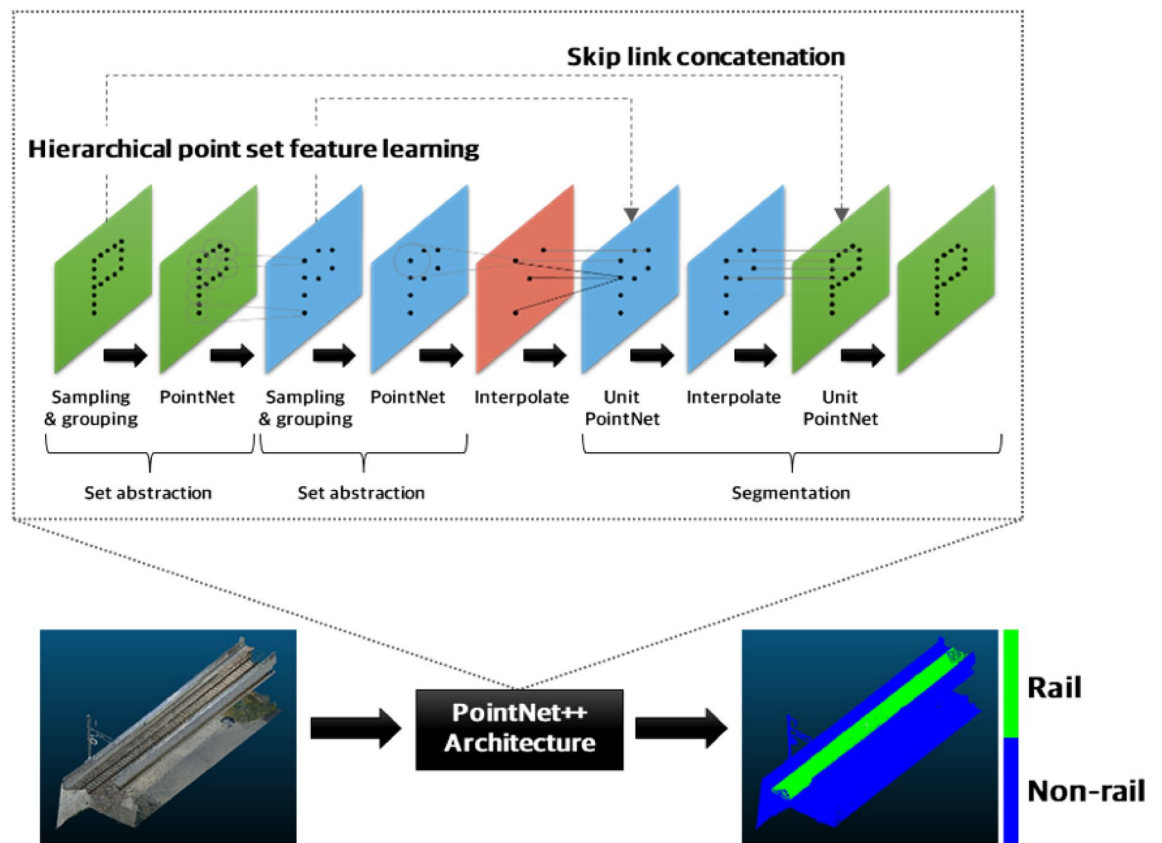


Fig. 1. Network architecture for PointNet++.

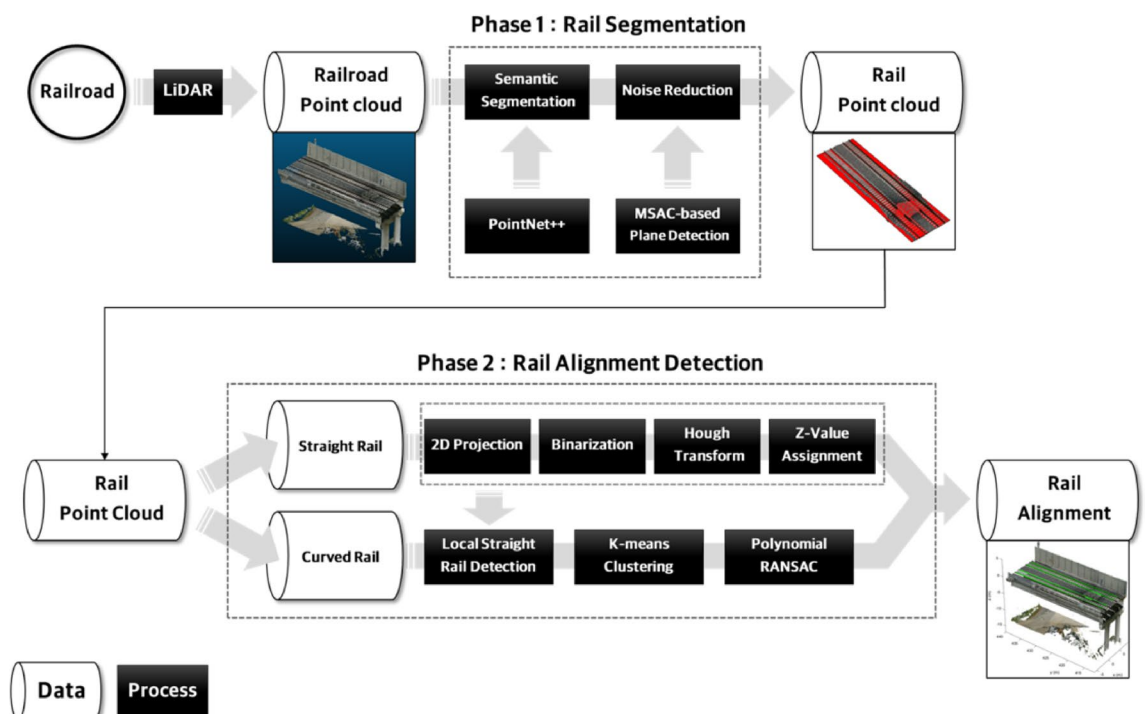


Fig. 2. Overview of the proposed automatic 3D railroad alignment detection method.

Phase 1: rail segmentation

Figure 2 presents an overview of Phase 1. The first step in detecting rail alignment, as proposed in this study, is to scan the railroad and acquire point cloud data, typically using LiDAR. However, the collected data contain non-railroad elements and noise, requiring post-processing to isolate the railroad components. To overcome this challenge, a deep learning-based 3D semantic segmentation approach is proposed to automate the process. We use PointNet++ to segment railroad components from LiDAR data, employing transfer learning to adapt the model for this task. Labels are categorized into rail components and non-rail components. Further details on the network configuration are provided in Chap. 4.

The segmented point cloud data still contains some non-rail components and noise due to the limitations of the deep learning model. To address this, the proposed method incorporates an MSAC-based plane detection algorithm as a crucial post-processing step to further refine the data and ensure reliable rail extraction. MSAC (M-estimator Sample Consensus) is an enhanced version of the Random Sample Consensus (RANSAC) algorithm³⁵, which operates by randomly selecting sample data and determining consensus. Unlike the standard RANSAC, MSAC improves robustness by minimizing the squared error of inliers, thereby yielding more accurate plane fitting results. In this study, the MSAC model was configured specifically to detect planes associated with the rail, making it well suited for distinguishing rail structures from surrounding noise.

The MSAC-based rail detection process consists of four steps. First, three random points in the point cloud are sampled to establish the initial model (Fig. 3a). Second, a planar model is constructed from these sampled points (Fig. 3b). Third, a threshold value is set based on the distance from the planar model. Finally, inliers and outliers are segregated, with points within the threshold classified as inliers (Fig. 3c). This process is iteratively repeated until the maximum number of inliers is identified. Ultimately, the plane corresponding to the rail is extracted, effectively removing noise and irrelevant points. Because rails generally form a planar structure atop the railbed, the MSAC-based approach allows precise localization of the rail while simultaneously suppressing noise, highlighting its essential role in enhancing the robustness of the proposed method.

Phase 2: rail alignment detection

After obtaining the segmented rail point cloud, the next step involves detecting the alignment. Figure 2 offers an overview of the proposed Phase 2. In this study, the Hough transform³⁶ is applied to discern the rail alignment from the rail point cloud model acquired in Phase 1. Typically utilized to detect specific shapes, such as straight lines and circles within a 2D image, the Hough transform is adapted in this research to identify a straight line (i.e., rail alignment) within point cloud data, rather than a 2D image. This adaptation introduces a “rail alignment detection” method, outlined as follows.

Initially, to minimize computational costs and memory demands, dimension reduction is implemented. The dimensionality reduction process of the 3D point cloud is as shown in Fig. 4. The Z-axis in the 3D point cloud data, captured by LiDAR, is perpendicular to the ground surface. Through the previous process, the plane of the ground surface was estimated (Fig. 4a). The data loss is minimal even when reducing the Z dimension of a 3D point cloud that exists on a plane. As shown in Fig. 4b, the Z value is removed to convert it into a 2D point cloud. The Z-coordinate information is preserved for use in later stages. Following this, the 2D point cloud data is binarized and dimensionally reduced as shown in Fig. 4c. Binarized data can be subjected to the Hough Transform.

To detect straight lines in binarized data using the Hough Transform, the process starts by mapping each data point that represents an edge (typically marked as 1 in binarized data) to a series of lines in the Hough transform matrix, which is defined by parameters like angle and distance from the origin. Each of these lines in the Hough transform matrix corresponds to a potential line in the original data space that could pass through the original point. The Hough Transform accumulates votes in a 2D array or accumulator, where each cell represents a specific combination of distance and angle parameters. When multiple data points contribute to the same cell in this accumulator, it indicates a strong candidate for a line in the data space, with the cell's value representing the number of points lying on that line. By setting a threshold on these values, significant lines that have a substantial

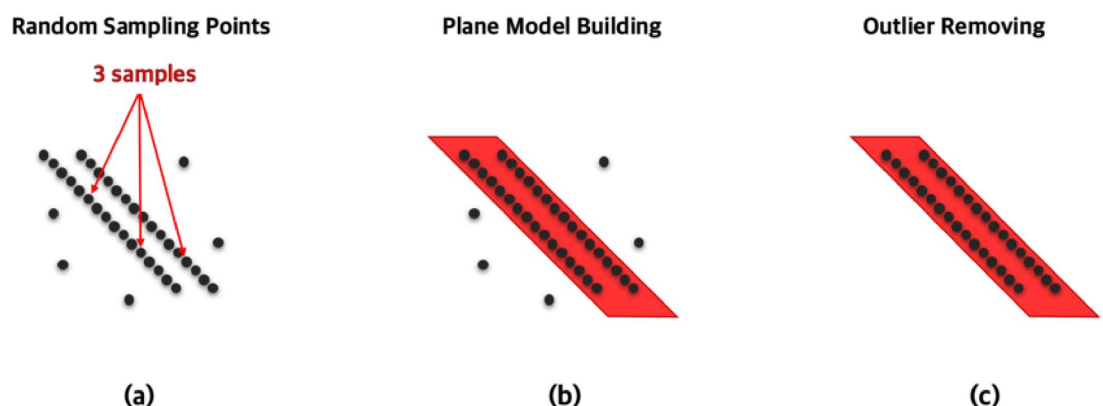


Fig. 3. Procedure for applying MSAC to reduce noises in 3D point cloud: **a** Random sampling of three points. **b** Building a plane model from point cloud data. **c** Point cloud data with outliers removed.

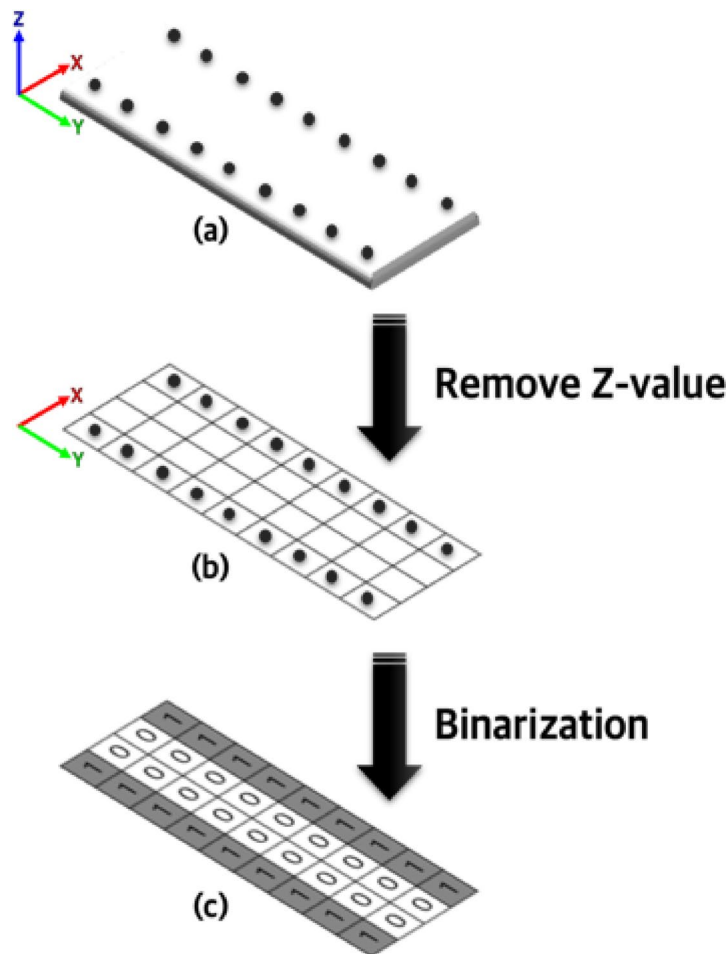


Fig. 4. Dimensionality reduction process of the point cloud.

number of points can be identified and extracted from the binarized data, allowing for the detection of linear patterns even in noisy or incomplete datasets.

The proposed method for rail alignment detection effectively automates the identification of rail alignment but is initially limited to straight rail sections. When applied to curved rails, the method may fail to accurately capture the alignment. To address this limitation, this study introduces an extended procedure specifically designed for curved rails, termed the curved rail alignment detection method.

The workflow of curved rail alignment detection is illustrated in Fig. 5. The fundamental principle of this approach is that a curve can be approximated by a sequence of small straight-line segments. To implement this, each curved rail is uniformly divided into subsections, and the alignment detection method for straight rails is applied to each segment. By increasing the number of subsections, the curvature of the rail can be represented with higher fidelity. The detection process is composed of three main steps: (1) local straight rail detection for each subsection, (2) Rail-wise merging using k-means clustering, and (3) polynomial RANSAC-based curve detection to refine and accurately reconstruct the rail alignment.

After detecting the alignment within each subsection, the results are aggregated for each rail. For this aggregation, K-means clustering³⁷ is employed due to its efficiency in grouping rail segments based on distance. To further refine the results, polynomial RANSAC is applied to correct errors that may arise during the Hough transform or clustering. Finally, curve fitting techniques are used to estimate the overall alignment of the curved rail.

This comprehensive strategy emphasizes the accurate extraction of curved rail geometries, overcoming the limitations of straight-line detection and enabling robust alignment estimation even in complex curved sections.

Experiments

Dataset

To validate the proposed method, an on-site experiment was conducted at the Osong railroad test track in Nojang-ri, Jeondong-myeon, Sejong-si, Republic of Korea. Five bridges on the Osong test track were selected for this evaluation: the 1st, 2nd, 3rd, 5th, and 6th bridges (Fig. 6a). Notably, the 5th bridge is the only double-track bridge among them, while the others are single-track bridges. Point cloud data for these bridges were acquired using the RIEGL VZ-400i LiDAR (Fig. 6b). The LiDAR-acquired data was post-processed, and

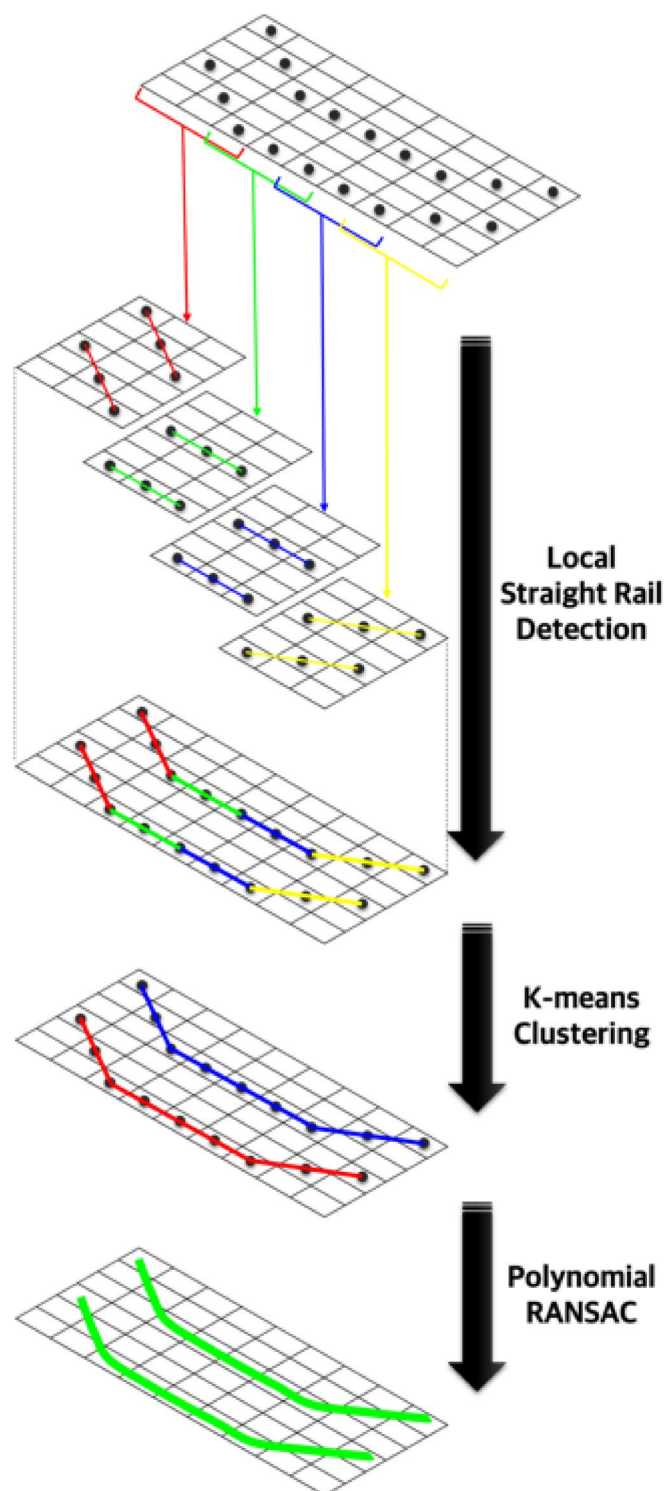


Fig. 5. Flowchart of curved rail alignment detection.

detailed information for each dataset is provided in Table 1. The specifications of the RIEGL VZ-400i LiDAR are presented in Table 2.

Results of rail segmentation

Initially, the rail segmentation method (Phase 1) was implemented to remove extraneous data from the point cloud. The hyperparameters used for training with the PointNet++ architecture are outlined in Table 3. Training was conducted with a batch size of 16 and over a total of 300 epochs, employing the Adam optimizer and a learning rate of 0.005. For training, spans from the 2nd bridge (8 spans) and the 6th bridge (1 span) were used,



(a)



(b)

Fig. 6. **a** Image of the 3rd bridge on the Osong test track. **b** Image of the RIEGL VZ-400i.

Name of the bridge	Rail type	Number of spans	Total length (m)
1st	Single-track	2	68.8
2nd	Single-track	8	246.5
3rd	Single-track	2	52.8
5th	Double-track	13	420
6th	Single-track	1	41.7

Table 1. Number of point clouds used for the experiments.

Parameters	Specifications
Laser acquisition speed(max)	122,000 meas/sec
Data acquisition distance	600 m
Accuracy	5 mm
Wide field of view	100° × 360°

Table 2. Specifications of RIEGL VZ-400i.

Hyperparameters	
Batch size	16
Epoch	300
Optimizer	Adam optimizer
Initial learn rate	0.0005
Learn rate drop period	10
Learn rate drop factor	0.1
Gradient decay factor	0.9

Table 3. Hyperparameters used for the experiments.

while data not involved in the training process (i.e., from the 1st, 3rd, and 5th bridges) was allocated as testing data. In addition, to overcome the limitations of the relatively small dataset, we further incorporated synthetic data into the training set.

The proposed method achieved an average accuracy of 0.9763 and an average Intersection over Union (IoU) of 0.7958 for rail segmentation, as reported in Table 4. Fig. 7a showcases an example of utilizing PointNet++ with transfer learning. Despite the effective segmentation, some residual noise remained due to the intrinsic limitations of the deep learning model. To address this noise, an MSAC-based noise reduction technique was

Classes	Accuracy	IoU
Rail	0.9763	0.6784
Non-rail	0.9521	0.9132
Average	0.9763	0.7958

Table 4. Result for the Rail Segmentation (Osong Railroad 1st, 3rd, 5th Bridges and Synthetic data).

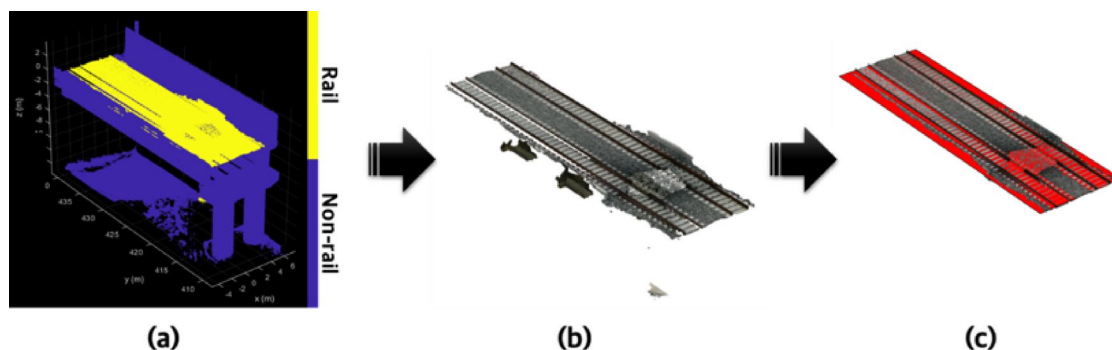


Fig. 7. **a** Semantic segmentation result obtained using transfer-learned PointNet++, **b** result of the rail class containing noise, **c** plane detected after noise removal using MSAC.

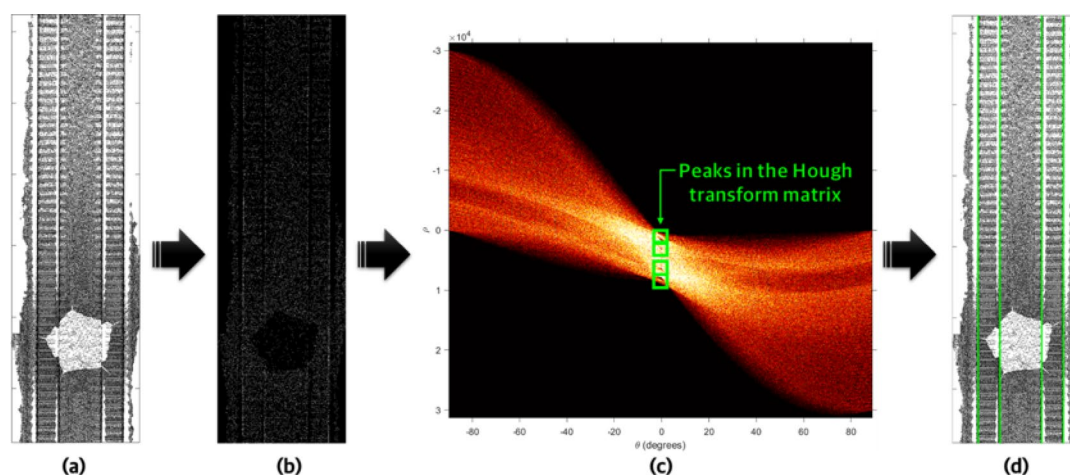


Fig. 8. **a** Point cloud on the MSAC plane, **b** binarized point cloud, **c** peaks (green box) in the Hough transform matrix, **d** result of back-projected the peaks in the Hough transform matrix.

employed. Fig. 7b shows a railroad point cloud with noise prior to the application of MSAC, whereas Fig. 7c demonstrates the railroad point cloud post noise removal using MSAC.

Result of rail alignment detection

Following the initial phase, the proposed rail alignment detection method (Phase 2) was employed to determine the alignment of both straight and curved rails. To evaluate the accuracy of the proposed method, reference alignment data were obtained by recording the 3D point cloud of the rail at 1-meter intervals.

To validate the rail alignment detection method for straight rails, experiments were conducted on the 3rd and 5th bridges of the Osong railroad, both characterized by straight rail sections. The application of the Hough transform required preprocessing of the 3D point cloud, which involved converting it into a 2D point cloud using a 2D projection method (see Fig. 8a, b). The 2D point cloud was then binarized to represent it within the Hough transform matrix (Fig. 8c). By applying the Hough transform, the peak point in the matrix was identified and translated into a straight-line representation (Fig. 8d). Incorporating the corresponding Z values enabled rendering of the straight line in 3D space, thereby accurately depicting the rail alignment.

To validate the rail alignment detection method for curved rails, an experiment was conducted on the Osong railroad's 1st bridge, which features curved rail sections (Fig. 9). The rail point cloud was segmented at 1-meter intervals, and alignment detection was performed for each segment following the same procedure

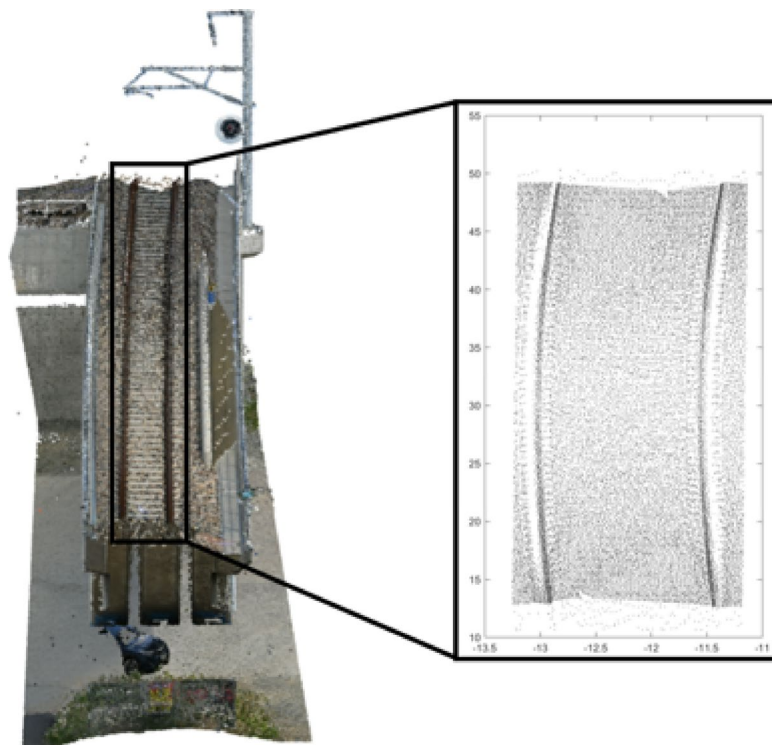


Fig. 9. Point cloud of Osong Railroad 1st bridge with curved rails.

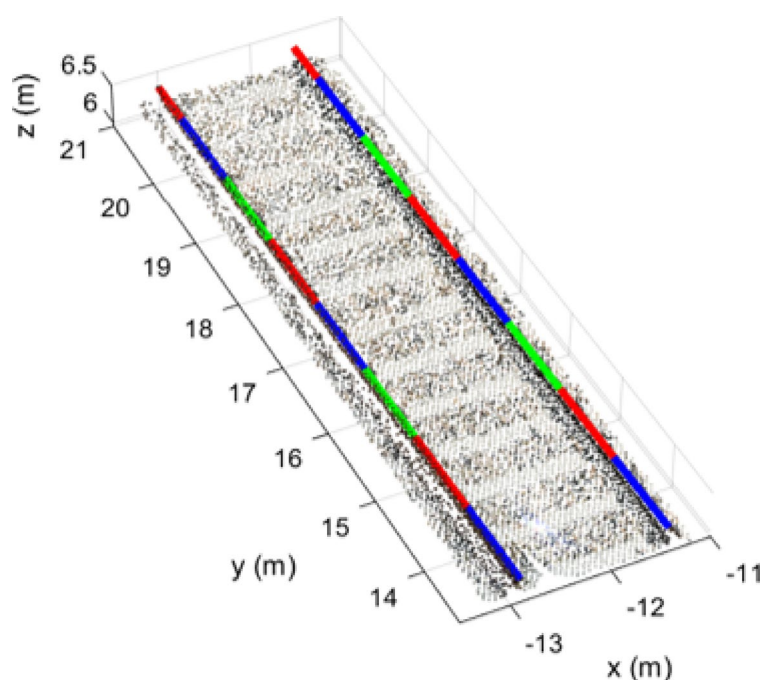


Fig. 10. Curved rail alignment detected using the proposed method at 1-meter intervals.

used for straight rails. The outcomes of the proposed method are illustrated in Fig. 10. Because the detected rail alignments are initially represented as separate linear entities, k-means clustering was employed to merge them, followed by RANSAC-based curve fitting to minimize errors and reconstruct the continuous curvature. Figure 11a shows the final alignment obtained using the proposed method, while Fig. 11b presents a comparison between the RANSAC-based alignments results and the proposed method in coordinate space.

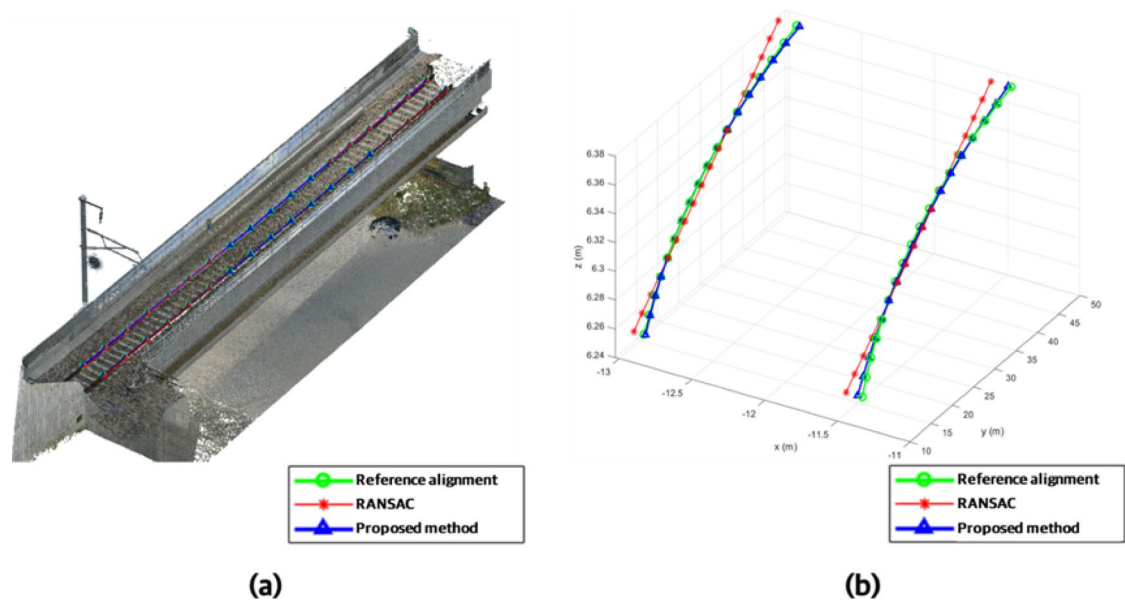


Fig. 11. **a** Result of the rail alignment detection for Osong Railroad 1st bridge (curved rail), **b** Result of the rail alignment in coordinate space.

Method	RMSE (mm)		
	Rail of 1st bridge	Rail of 3rd bridge	Rail of 5th bridge
RANSAC	9.64	7.47	7.66
Proposed Method	3.89	3.12	3.71

Table 5. Result for the rail alignment detection osong railroad bridges.

Table 5 reports the Root Mean Square Error (RMSE) results obtained from applying the proposed method to the rails of the Osong railroad’s 3rd and 5th bridges. The average RMSE was 3.415 mm, demonstrating the effectiveness of the proposed method in detecting the alignment of straight rails. Moreover, when compared with the conventional RANSAC-based approach, the proposed method achieved higher accuracy, confirming its superior performance in maintaining alignment precision even in complex rail geometries.

Discussion

There are three primary sources of error identified in this study. The first arises during data collection with LiDAR. Such errors can be broadly categorized into (1) errors caused by diffuse reflection of light and (2) errors due to imperfections in the point cloud registration process. While these issues are largely unavoidable, ongoing advancements in LiDAR technology are expected to mitigate their impact in future applications.

The third source of error pertains to curved rails. Although the proposed method improves the accuracy of alignment detection for curved sections, it cannot fully eliminate errors introduced by rail curvature. Excessive subdivision of curves may lead to subsections containing an insufficient number of points to reliably approximate the curvature. To reduce this source of error, acquiring higher-density point cloud data is essential, and incorporating curve-aware deep learning models may further enhance robustness.

Taken together, these findings highlight the current limitations of the proposed method while also pointing to clear avenues for future research, including improved sensing hardware, enhanced segmentation models, and curve-specific alignment detection strategies.

Conclusion

This study presented a fully automated method for detecting 3D railroad alignment from LiDAR-based point cloud data, structured into two main phases: (1) rail segmentation and (2) rail alignment detection. The first phase involved the use of a deep learning-based semantic segmentation model—specifically, PointNet++—to extract only the relevant rail components from raw point cloud data, which typically contains a variety of non-rail structures and background noise. By employing transfer learning, the network was effectively adapted to distinguish between rail and non-rail elements, thereby enabling robust preprocessing across diverse railroad environments. The second phase focused on the precise detection of railroad alignment, which was further divided into straight and curved alignment detection processes. For curved segments, a key innovation of this study was the application of local linear approximation, in which the rail was divided into multiple uniformly spaced sub-segments. The alignment within each sub-segment was estimated using the same method applied to

straight segments, and the resulting data were then aggregated and refined through clustering and polynomial fitting techniques.

To validate the effectiveness and practical applicability of the proposed method, an on-site field test was conducted at the Osong railroad test track, located in Nojang-ri, Jeondong-myeon, Sejong-si, Republic of Korea. Five bridges were selected as the testbed, representing a combination of single- and double-track rail sections. The proposed method demonstrated strong performance, achieving an average RMSE of 3.57 mm in 3D rail alignment estimation. This level of precision indicates that the method is not only accurate but also reliable under real-world conditions.

Compared to conventional manual inspection methods, which are both labor-intensive and time-consuming, the proposed approach offers significant improvements in terms of efficiency and scalability. In the long term, this method holds great potential for enabling continuous, large-scale monitoring of railroad infrastructure as part of intelligent transportation and infrastructure management systems.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Received: 31 March 2025; Accepted: 3 December 2025

Published online: 13 December 2025

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Acknowledgements

This work was supported by Chungbuk National University NUDP program (2025) and National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (RS-2022-NR071667).

Author contributions

J.L. managed the overall research progress and wrote the main manuscript. J.P. carried out data collection and post-processing. H.S. conducted a review of the methodology. H.Y. directed and supervised the overall research. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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